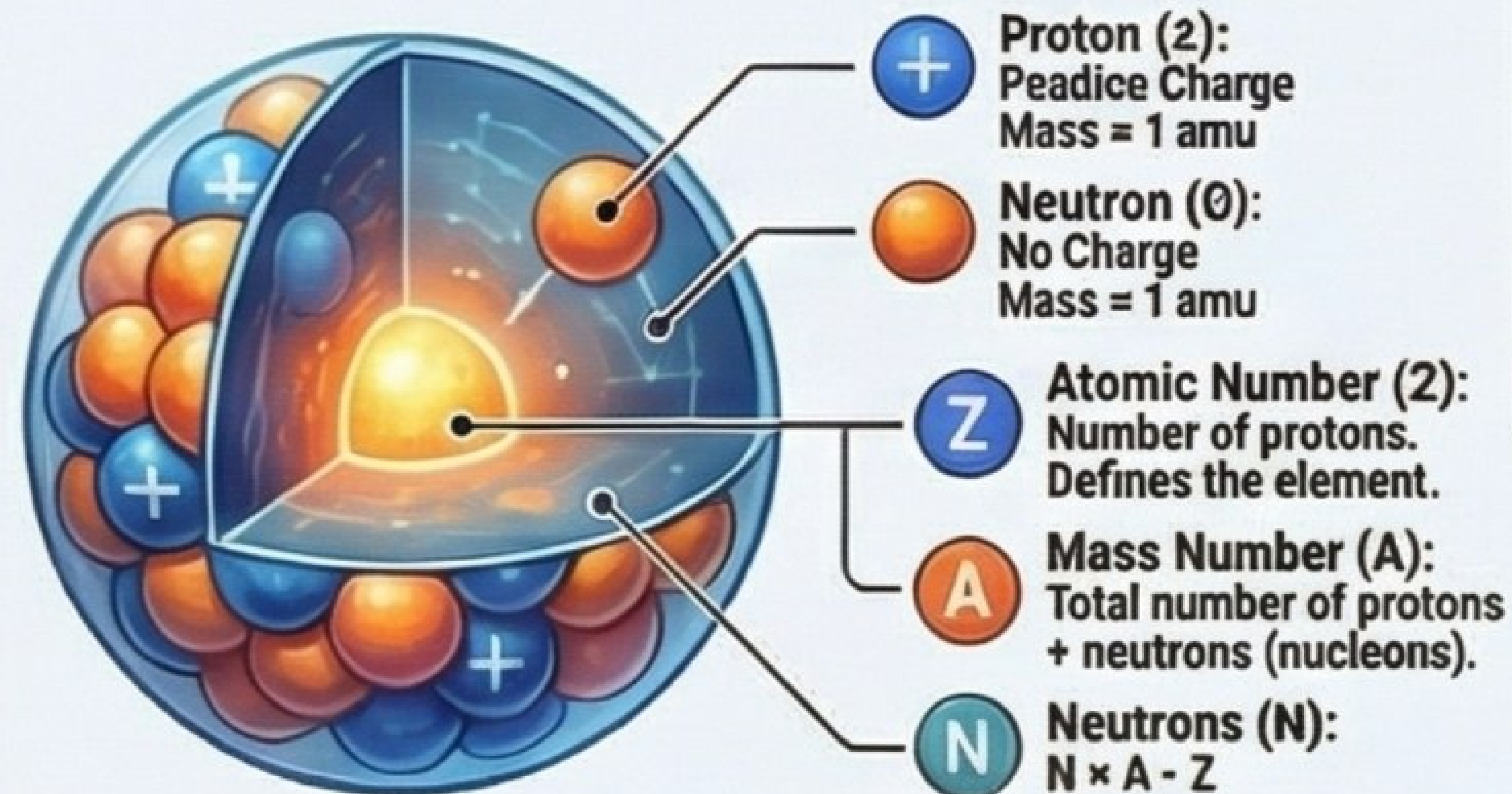


The Anatomy of a Nucleus



The Nucleus: The Atom's Core

Control part, concentrated protons & neutrons, containing all positive charge and almost all mass.

Decoding Uranium-235 ($^{235}_{92}\text{U}$)

$^{92}_{92}\text{U}$ — $Z = 92$ (82 protons)
 $^{92}_{92}\text{U}$ — $A = 235$ (Total nucleons)
 $N = 235 - 92 = 143$ neutrons.

Classifying Nuclei

Isotopes



Same protons (Z), different neutrons.

Isotones



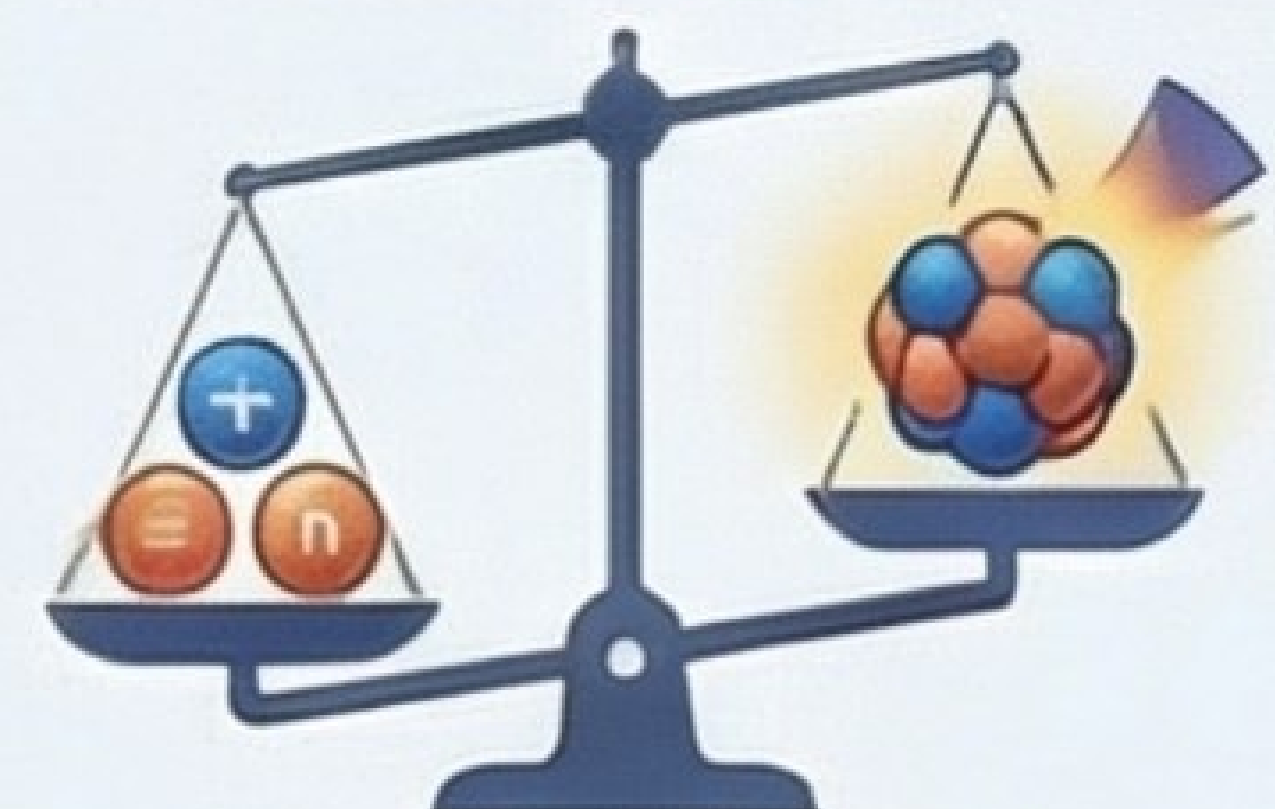
Some neutrons, different protons.

Isobars



Same mass number (A), different protons & neutrons.

Mass, Energy, and Stability



Mass Defect (Δm)

Nucleus mass < Total mass of individual nucleons. "Missing" mass converts to energy.

$$E = mc^2$$

Binding Energy (BE):

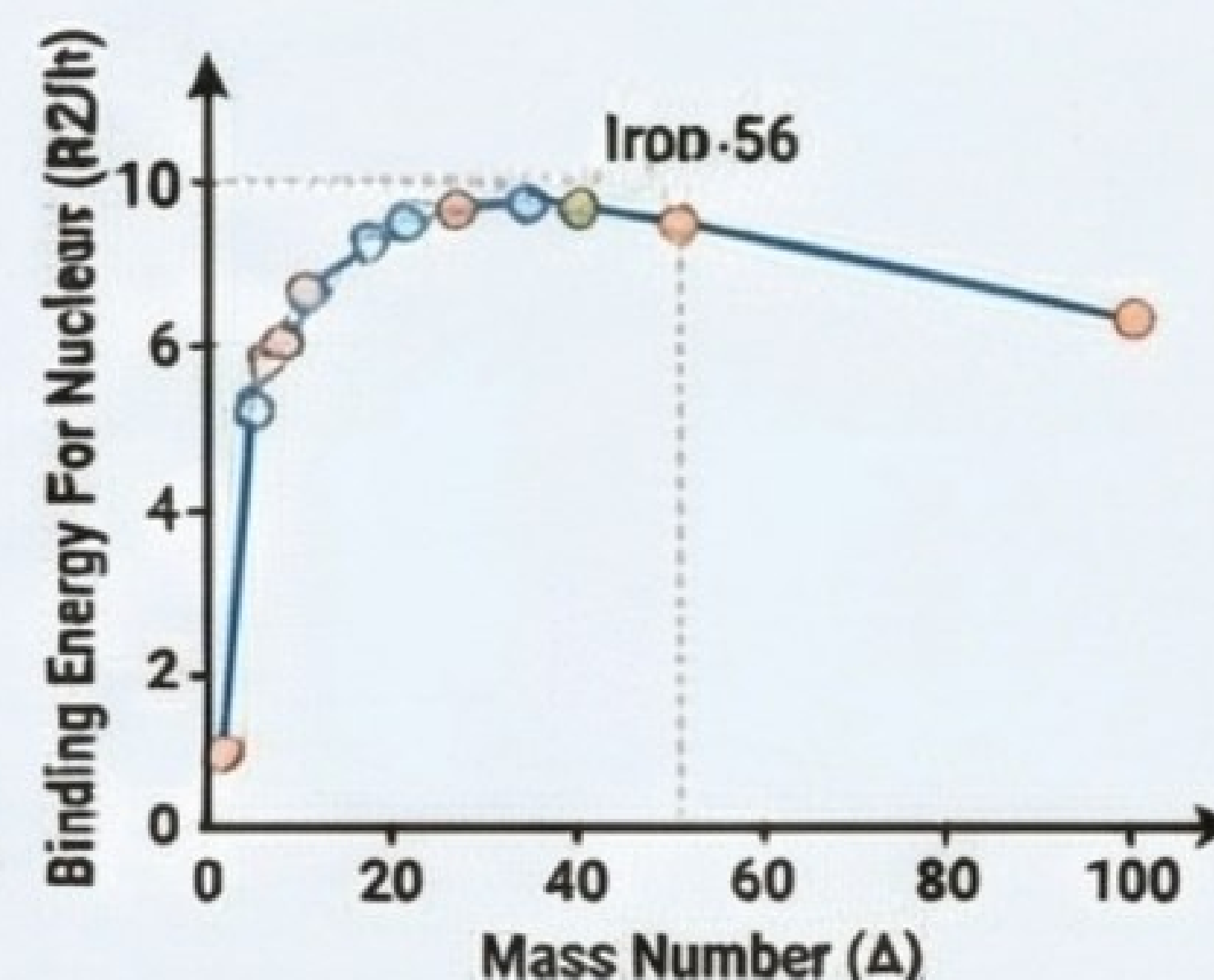
Energy released when nucleus forms.
 Energy equivalent of mass defect.

The Magic Number:

1 amu = 931 MeV BE.

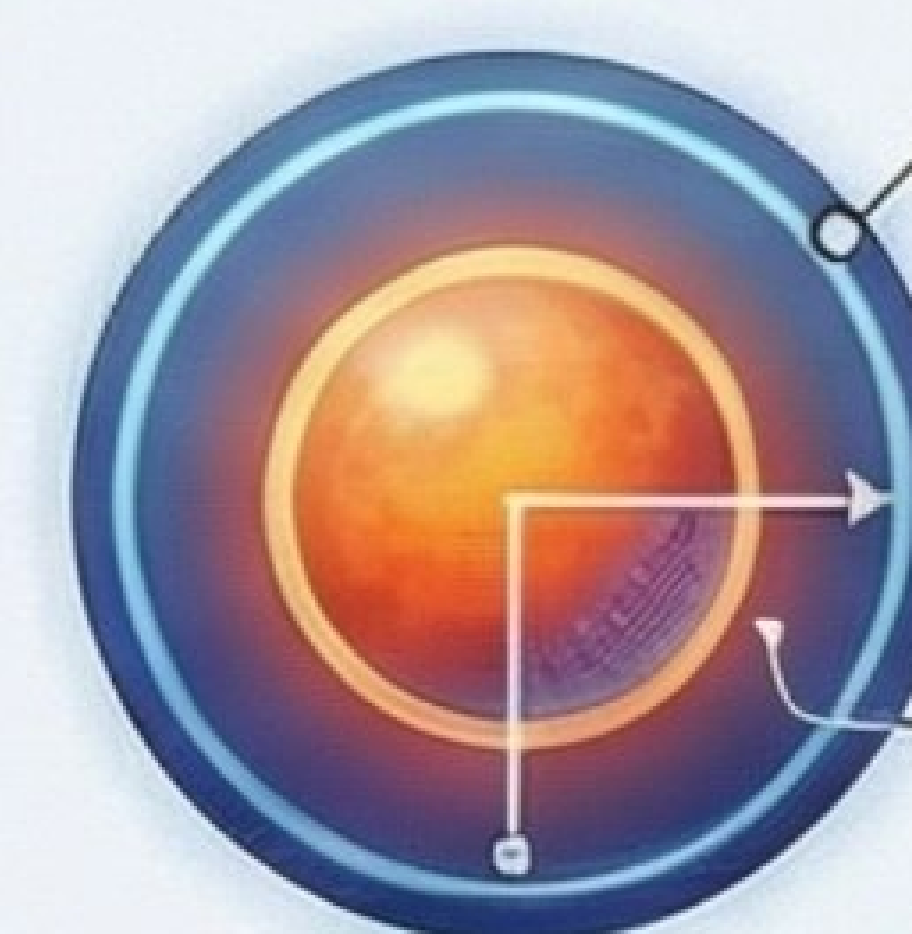
Higher BE/A =
 More stable nucleus.

Iron-56 (Fe) is most stable (~0.6 MeV).



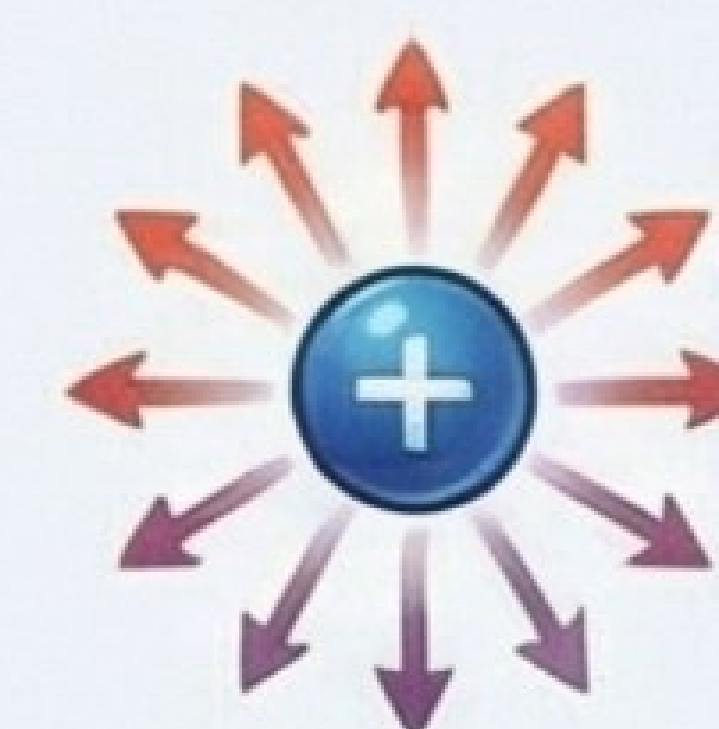
A Quick Guide to the Atomic Nucleus

Key Properties & Measurements



Universal Nuclear Density:
 $\sim 2.29 \times 10^{17} \text{ kg/m}^3$
 (Constant for all atoms, independent of size).

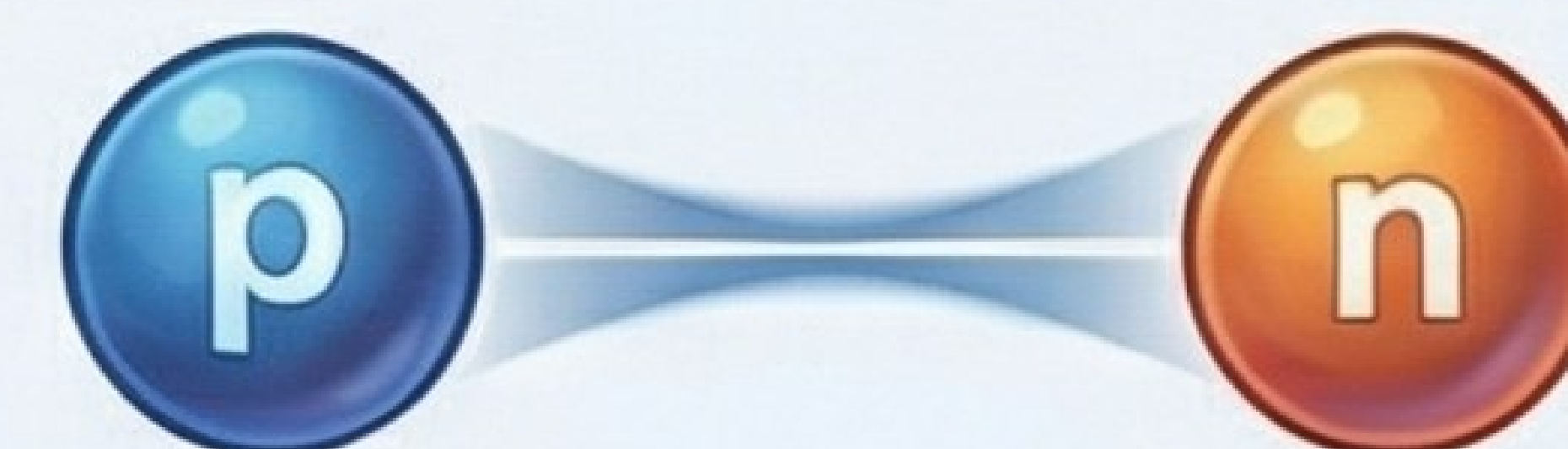
Nuclear Radius (R) = $R_0 A^{1/3}$
 ($R_0 = 1.2$ Fermi or $1.2 \times 10^{-10} \text{ m}$).



Nuclear Charge:
 Total positive charge = $+Ze$.

Atomic Mass Unit (amu or u):
 Standard unit, 1/12th mass
 of Carbon-12
 (1 amu = $1.66 \times 10^{-27} \text{ kg}$).

The Force That Binds



The Strong Nuclear Force

Powerful attractive force holding nucleons together, overcoming electrostatic repulsion.

Strongest Force in Nature

Charge Independent
 (acts equally
 p-p, n-n, p-n)

Extremely Short Range
 (effective ≈ 10 Fermi,
 rapidly $\propto 1/r^2$ & $1/r^3$).

Nuclear Reactions: Fission vs. Fusion



Process	Heavy nucleus splits into lighter ones	Two light nuclei combine to form heavier one
Trigger	Heavy nucleus absorbs slow neutron	Extremely high temperatures & pressure
Example	Uranium-235 splitting	Hydrogen isotopes fusing to Helium (Sun)
Energy Release	Huge (e.g., ~200 MeV per U-235)	Even more per nucleon than fission
Byproducts	More neutrons, can cause chain reaction	Neutrons or other particles
Application	Nuclear power plants, atomic bombs	Stars, experimental reactors (ITER)